

Comparative Analysis of YOLOv10, YOLOv11, and YOLOv12 for Retail Self-Checkout Systems

Barla Sai Vardhan
Dept. of ECE Vardhaman
College of Engineering
Shamshabad, India
barlasaivardhan22ec@student.vardhaman.org

Godha Karthikeyan Yadav
Dept. of ECE Vardhaman
College of Engineering
Shamshabad, India
godhakarthikeyanyadav22ec@student.vardhaman.org

K.H.Murali
Dept. of ECE Vardhaman
College of Engineering
Shamshabad, India
k.murali@vardhaman.org

Nallela Sai Nikhil Tej
Dept. of ECE Vardhaman
College of Engineering
Shamshabad, India
nallelasainikhiltej22ec@student.vardhaman.org

Abstract—Automated self-checkout systems are pivotal in the modernization of the retail sector, yet their effectiveness hinges on the underlying technology’s ability to recognize products with both speed and precision. This paper explores a deep learning framework designed for this application, leveraging the You Only Look Once (YOLO) object detection algorithm, distinguished by its single-stage architecture. Unlike traditional two-stage detectors that require a separate process to first identify regions of interest, YOLO frames object detection as a single regression problem, directly predicting bounding boxes and class probabilities from full images in one evaluation. This unified approach significantly reduces computational complexity, making it ideal for real-time applications. We conduct a rigorous comparative analysis across three successive generations of this algorithm—YOLOv10, YOLOv11, and YOLOv12—and their respective nano (n), small (s), and medium (m) variants. Performance was benchmarked using mean Average Precision (mAP) and latency to measure speed. Findings show YOLOv12m achieved highest precision with mAP@50 of 98.1% at 35.65 ms latency, while YOLOv11n offered a balanced trade-off with 97.8% mAP@50 and 17.29 ms latency.

Index Terms—Automated self-checkout, Object detection, YOLO, Deep learning, Real-time systems, Retail automation, Image Processing

I. INTRODUCTION

Object detection has become a cornerstone in modern computer vision, with applications spanning autonomous driving, surveillance, industrial automation, and retail checkout systems. Traditional two-stage detectors such as Faster R-CNN have demonstrated high accuracy but often suffer from computational inefficiencies, limiting their applicability in real-time environments [1]. To address these limitations, recent research has focused on improving single-stage detectors and hybrid approaches that balance accuracy with efficiency.

Early advancements introduced multi-region and segmentation-aware convolutional models, which enhanced object localization by incorporating semantic cues [2].

Subsequently, single-shot detectors such as the SSD model offered faster inference compared to two-stage approaches, paving the way for real-time implementations [3]. These developments laid the foundation for applying deep learning-based detection in emerging fields, including automated retail checkout. Region-based methods have been explored for reducing misclassification during product recognition in retail environments, highlighting the potential of deep detection networks in this domain [4].

The demand for cashier-less checkout systems continues to grow as retail industries adopt automation to reduce human intervention. Recent efforts have leveraged CNN-based solutions to design end-to-end deep learning models capable of recognizing and processing multiple products efficiently [5]. Among such approaches, robust frameworks like DeepACO have integrated deep learning with task-specific optimizations to improve checkout system reliability [6].

Building upon these developments, the YOLO (You Only Look Once) series has emerged as a dominant real-time detection framework. With continuous evolutionary improvements, YOLO-based architectures have demonstrated superior balance between inference speed and detection accuracy, particularly in dense and dynamic environments such as retail checkout [7]. Recent analyses have highlighted the broad applications of YOLO in domains like autonomous driving, validating its versatility and adaptability [8]. Additionally, comprehensive reviews confirm YOLO’s significant role in shaping current real-time object detection systems by benchmarking its variants against other state-of-the-art methods [9].

As the YOLO framework progresses, its latest iterations, including YOLOv12, have introduced architectural enhancements such as attention-centric mechanisms designed to further boost performance in real-time applications [10]. Such advancements directly benefit domains where rapid and precise detection is critical, including retail self-checkout systems. At

the same time, domain-specific adaptations of YOLO, such as its use in electrical equipment classification, demonstrate how tailored versions of these models can optimize detection tasks for specialized environments [11]. A recent decade-long review of the YOLO series reaffirms its dominance and the continuous improvements that make newer variants highly suitable for industrial applications [12].

In this work, we present a comparative analysis of YOLOv10, YOLOv11, and YOLOv12 in the context of real-time retail self-checkout systems. By evaluating accuracy, robustness, and computational efficiency, this study aims to provide insights into the strengths and trade-offs of recent YOLO versions, thereby contributing to the development of more seamless and reliable cashier-less retail checkout technologies.

II. RELATED WORK

Automated retail checkout systems have attracted substantial research interest due to their potential to revolutionize the retail industry by reducing human intervention and checkout times. Several deep learning-based approaches have been proposed to improve product detection and recognition accuracy in real-time scenarios.

Shoman et al. [4] proposed a region-based deep learning method focusing specifically on automated retail checkout applications. By leveraging region proposal networks combined with deep feature extractors, their system improved localization accuracy; however, computational overheads made real-time deployment difficult.

Lee et al. [5] developed a cashierless checkout vision system employing deep learning techniques designed for smart retail settings. Although effective in controlled environments, their system struggled with dynamic lighting conditions and product variations common in real-world stores, highlighting the need for more adaptable detection frameworks. Recent advancements in YOLO-based models have shown promising results for real-time object detection in retail. Pham et al. [6] introduced DeepACO, a robust deep learning-based automatic checkout system that integrates convolutional neural networks with advanced optimization techniques. Their approach demonstrated improved reliability in identifying multiple products simultaneously but still faced challenges with occlusion and visually similar products, limiting scalability in complex retail environments. Qiu et al. [7] proposed LD-YOLOv10, a lightweight target detection algorithm based on YOLOv10, focusing on reducing model complexity while maintaining high detection accuracy. Although the approach demonstrated improved efficiency and suitability for resource-constrained platforms, challenges persisted in accurately detecting densely packed objects and distinguishing visually similar targets. Gao et al. [11] explored improvements of YOLOv10 architecture for specific applications like electrical equipment classification, showing how model fine-tuning and domain adaptation can enhance detection performance. Their findings suggest similar potential for retail checkout systems but underscore the

need for careful dataset curation and augmentation to mitigate false positives.

Comprehensive reviews by As'arry et al. [8] and Sapkota et al. [12] analyzed the evolution of YOLO algorithms, emphasizing their suitability for applications requiring fast inference such as retail checkout. These studies identified that while YOLO models excel in speed, accuracy trade-offs persist, particularly for small or occluded objects frequently encountered on retail shelves. He et al. [13] conducted a comprehensive performance evaluation of YOLOv10, YOLOv11, and YOLOv12 models across multiple benchmark datasets. Their experiments show YOLOv12 achieves highest precision and mean Average Precision (mAP@50) with scores up to 0.931, closely followed by YOLOv11 variants at 0.926-0.933, while YOLOv10 lags slightly behind with peak mAP values near 0.921. YOLOv11n achieved the highest inference speed (2.4 ms), outperforming YOLOv10 and YOLOv12 nano models, making it well-suited for latency-sensitive real-time applications. However, YOLOv12's attention-centric architecture, while providing the best accuracy, incurs slightly higher computational overhead, which can be limiting on resource-constrained devices.

Khan and Jensen [14] provide an architectural overview of YOLOv11, highlighting novel modules such as the C3k2 block and the C2fPSA attention mechanism that enhance feature extraction and spatial reasoning. These innovations enable improved detection of small and occluded objects, which is critical in cluttered visual environments. The study further notes that such efficiency gains are particularly advantageous for embedded platforms, although challenges remain in further reducing model size without compromising detection accuracy.

Vats and Anastasiu [15] presented an enhanced retail checkout framework that integrates video inpainting with YOLOv8-based object detection and DeepSort tracking to address challenges caused by occlusion and object overlap in checkout environments. The proposed system employs video inpainting to reconstruct partially occluded product regions, enabling more reliable detection and tracking across video frames. Experimental results demonstrate improved detection continuity and reduced identity switches compared to conventional frame-based detection approaches.

Hameed et al. [17] proposed a real-time retail theft classification system using the YOLO object detection framework. The study demonstrated that YOLO-based detectors effectively handle complex retail environments with dynamic backgrounds and varying object scales while maintaining low inference latency. These findings support the suitability of advanced YOLO variants, such as YOLOv11, for real-time retail applications requiring improved feature extraction and efficiency.

III. METHODOLOGY

This section details the methodology adopted for analyzing and benchmarking multiple YOLO architectures applied to automated self-checkout systems. The focus is on leveraging

cutting-edge, single-stage, real-time object detection frameworks specialized in fine-grained object localization and classification, which are critical for retail automation.

A. Dataset Description and Input Features

The dataset used in this study is the *Retail Store India New 2* dataset sourced from the Roboflow Universe [16]. It comprises 699 images, each annotated with bounding boxes around individual retail products. The dataset includes 42 product classes featuring a wide variety of common Indian retail items such as Amul Butter, Bisleri Club Soda, Cadbury Bournvita, Nescafe Classic Coffee, Tata Salt, Dove Beauty Cream, and more. The annotations reflect a diverse set of packaging types, sizes, and shapes, capturing the complexity of real store shelves.

Images were captured under varying environmental conditions, including diverse lighting conditions such as natural, fluorescent, and shadowed illumination; different camera angles (frontal, angled, partial views). The input for all YOLO models is an RGB image of size 640 x 640 pixels. An RGB image has three color representations, namely, red, green, and blue, which play a crucial role in distinguishing different types of retail products that resemble one another in terms of texture and color. The bounding box includes spatial features such as the x and y coordinates, width, and height of the products. The class label corresponds to the type of products, and the data augmentation includes fluctuations in the image, which can be used to handle changing illumination conditions as well as different types of occlusion that can occur in a real retail scenario.

The dataset is divided into training, validation, and test subsets in the ratios of 75%, 15%, and 10%, respectively. On average, each image contains multiple product instances, resulting in a moderately dense annotation set that is highly suitable for developing and benchmarking retail product detection and recognition systems. The dataset was captured using handheld cameras in realistic environments. It is also assumed that calibration errors for the cameras are minimal and that object boundaries are well represented by bounding box annotations. These assumptions enable consistent benchmarking of YOLO models under controlled yet realistic retail conditions.

B. YOLOv10 Architecture

YOLOv10 integrates several targeted architectural innovations that directly address key challenges in automated retail self-checkout systems [7]. The block diagram in Fig. 1 illustrates the model's modular design, comprising a backbone, neck, and multiple detection heads. Notably, Spatial-Channel Decoupled Downsampling (SCDDown) and C2f-Dual blocks in the backbone reduce redundancy and enable high-speed feature extraction, which is vital for real-time throughput on embedded devices common in retail environments. The inclusion of Efficient Multi-Scale Attention (EMA) modules in the neck improves semantic feature capture, supporting reliable detection under variable lighting and with occluded/stacked products.

The enhanced detection head—adopted from YOLOv8—enables precise object localization and reduces false positives, critical for dense shelf arrangements. A custom post-processing algorithm further refines bounding-box outputs in crowded and diverse retail scenes.

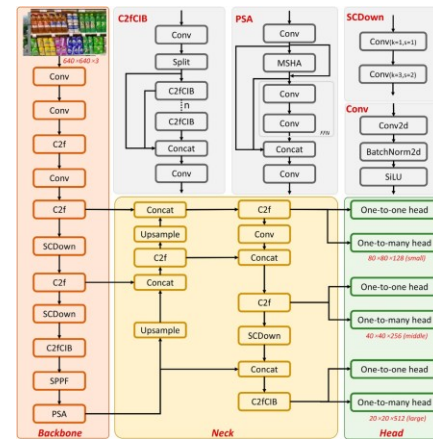


Fig. 1. Block diagram of the improved YOLOv10 architecture. Reproduced from [7].

C. YOLOv11: Efficiency-Oriented Enhancements

YOLOv11 introduces targeted enhancements that improve computational efficiency and detection accuracy for retail self-checkout applications [14]. As illustrated in Fig. 2, the architecture is organized into backbone, neck, and head modules. The backbone leverages stacked C3K2 blocks and deep convolutional layers for effective hierarchical feature extraction while maintaining low computational overhead, important for embedded point-of-sale devices.

Key blocks such as C3K2 (shortcut-enabled convolution) and SPFF enhance spatial feature aggregation, and C2PSA modules in the neck improve multi-scale attention. These elements allow YOLOv11 to capture detailed information from densely arranged and occluded retail products, optimizing detection precision in challenging scenarios. The head block applies mask-based outputs for refined localization.

YOLOv11 also adopts quantization, pruning, and improved squeeze-and-excitation for channel-wise feature recalibration. Together, these modifications ensure resource-efficient, high-throughput retail product recognition in real deployments.

D. YOLOv12: Attention-Centric Feature Refinement and Architectural Advancements

YOLOv12 marks the leading edge of YOLO series innovation, introducing architectural blocks and algorithmic enhancements tailored to address dense, cluttered retail shelf detection challenges [10]. As illustrated in Fig. 3, the backbone combines compound-scaling convolutions (balancing depth, width, and resolution) with the Residual ELAN (R-ELAN) module, enabling efficient, scalable feature extraction while controlling computation.

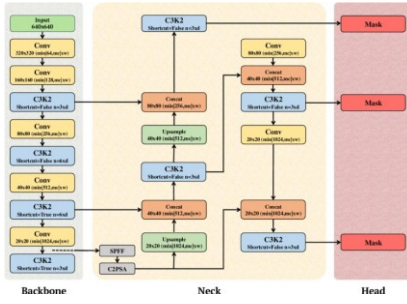


Fig. 2. YOLOv11 architecture with backbone, neck, and head modules. C3K2, SPFF, and C2PSA blocks support efficient feature extraction and semantic aggregation for retail product detection. Adapted from [14]

In the neck, Area Attention (A^2) and A2C2f blocks partition features for attention-guided aggregation, enhancing detection of small, overlapping products in retail views. A2C2f supports segmentation and localization, and upsampling layers facilitate multi-scale inference relevant for diverse retail product sizes. Normalization via Switchable Normalization (SN) improves network stability across batch domains, and coordinate attention injects positional cues that boost localization accuracy. The detection head integrates outputs from feature and segmentation heads, supporting robust bounding box and class predictions essential for accurate product recognition at checkout.

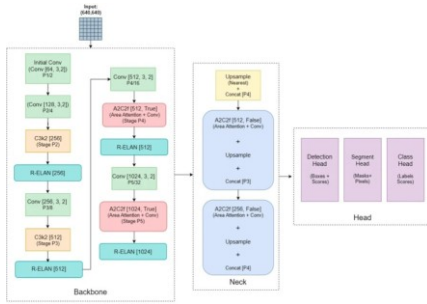


Fig. 3. YOLOv12: Architecture overview showing backbone, neck, and head

with relevant blocks for retail self-checkout.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Setup

The experiments were conducted using Google Colab Pro with the following hardware specifications:

- GPU: NVIDIA Tesla T4 with 16 GB VRAM
- CPU: 2 vCPUs
- RAM: 12 GB
- Disk: 100 GB cloud storage

The implementation was carried out in Python with the following major libraries and frameworks:

- Ultralytics YOLOv10, YOLOv11, YOLOv12 for object detection
- OpenCV for image preprocessing
- Pandas, NumPy, Matplotlib, Seaborn for data analysis and visualization
- PyTorch as the backend deep learning framework

B. Confusion Matrix and Metrics Calculation

Based on the provided confusion matrix values shown in Figure 4. for a representative model, we can calculate the key performance metrics: Accuracy, Precision, Recall, and F1-Score.

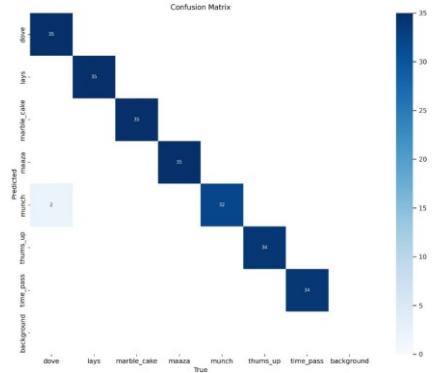


Fig. 4. Class-wise True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN) for a single model run.

To calculate the overall metrics for the entire model run, we must sum the values across all classes to get a total for TP, FP, FN, and TN.

- **Total True Positives (TP):** 35 + 35 + 35 + 35 + 32 + 34 + 34 = 240
- **Total False Positives (FP):** 0 + 0 + 0 + 2 + 0 + 0 + 0 = 2
- **Total False Negatives (FN):** 0 + 0 + 0 + 0 + 2 + 0 + 0 = 2
- **Total True Negatives (TN):** 207 + 207 + 207 + 205 + 208 + 208 + 208 = 1450

Using these totals, we can apply the standard formulas for each metric.

- **Accuracy:** Proportion of correct predictions out of total predictions. Accuracy is calculated using (1).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$= \frac{240 + 1450}{240 + 1450 + 2 + 2} = \frac{1690}{1694} \approx 0.9976 \quad (1)$$

This equates to an accuracy of approximately 99.76%, which indicates that a very high proportion of all predictions were correct.

- **Precision:** Proportion of correct positive predictions. Precision is defined using (2).

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{240}{240 + 2} = \frac{240}{242} \approx 0.9917 \quad (2)$$

This results in a precision of approximately 99.17%, showing that when the model predicts an item, it is correct over 99% of the time.

- **Recall (Sensitivity):** Proportion of actual positives that were correctly identified. Recall is given in (3)

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{240}{240 + 2} = \frac{240}{242} \approx 0.9917 \quad (3)$$

The recall is approximately 99.17%, indicating the model is highly effective at finding all of the actual items in the images.

- **F1-Score:** Harmonic mean of Precision and Recall. The F1-score is computed using (4).

$$\begin{aligned} \text{F1-Score} &= 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \\ &= 2 \cdot \frac{0.9917 \cdot 0.9917}{0.9917 + 0.9917} \\ &= 0.9917 \end{aligned} \quad (4)$$

The F1-score is 99.17%, suggesting a good balance between precision and recall.

C. Model Performance

The performance of the YOLO models was evaluated, and the key metrics are summarized in Table I.

TABLE I
PERFORMANCE METRICS OF YOLO MODELS (v10, v11, v12)

Model	Precision	Recall	mAP@50
YOLOv12-n	0.9311	0.9434	0.9732
YOLOv12-s	0.9591	0.9514	0.9794
YOLOv12-m	0.9482	0.9432	0.9813
YOLOv11-n	0.9357	0.9436	0.9783
YOLOv11-s	0.9545	0.9423	0.9791
YOLOv11-m	0.9470	0.9466	0.9733
YOLOv10-n	0.9110	0.8535	0.9444
YOLOv10-s	0.7049	0.7240	0.8153
YOLOv10-m	0.7684	0.7469	0.8588

The results summarized in Table I indicate that the **YOLOv12-m** model achieved the highest mAP@50 score of 0.9813, demonstrating superior detection accuracy at an Intersection over Union (IoU) threshold of 0.5. The YOLOv12-s and YOLOv11-s variants also achieved competitive performance with mAP@50 scores of 0.9794 and 0.9791, respectively, along with high precision and recall values.

However, detection accuracy alone does not fully determine suitability for real-time retail deployment. As shown in Table II, computational complexity and inference efficiency vary significantly across model variants. While YOLOv12-m provides the best accuracy, it incurs higher computational cost (67.92 GFLOPs) and increased latency (19.30 ms), resulting in lower inference speed (51.82 FPS). In contrast, YOLOv10-s achieves the highest inference speed of 83.36 FPS with comparatively lower latency (12.00 ms), making it advantageous for deployment scenarios with strict real-time constraints. These findings highlight the inherent trade-off between architectural complexity and inference efficiency, emphasizing that model selection for retail self-checkout systems must balance detection performance with hardware limitations and real-time processing requirements.

D. Performance Curve and Detection Analysis

The experimental results for YOLOv10, YOLOv11, and YOLOv12 on real-time automated retail self-checkout tasks

are illustrated through comprehensive training and validation curves, comparative mAP@50 performance plots, and precision-recall (PR) curves.

1. Loss and Metric Curves: The training process was analyzed by plotting epoch-wise curves for various metrics, as shown in Figure 5. The training and validation losses—including box loss, classification loss, and distribu-

tion focal loss (DFL)—show consistent convergence, which indicates effective model learning and proper generalization. The precision and recall metrics demonstrate a rapid increase

within the initial 20 epochs before plateauing above 0.95, confirming robust detection confidence and low rates of false positives and false negatives. Furthermore, the mAP@50 and mAP@50-95 metrics trend toward saturation near 1.0, which demonstrates high overall detection accuracy.

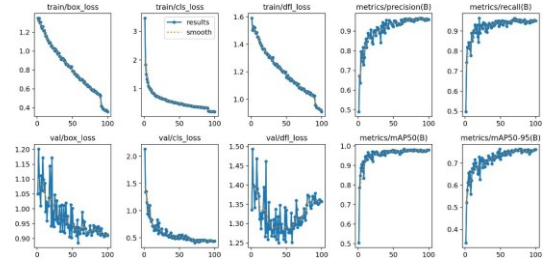


Fig. 5. Training and validation curves for multiple metrics during YOLO model training.

2. mAP@50 Comparative Performance: The comparative mAP@50 plot across YOLOv10 to YOLOv12 variants illustrates steep mAP increases in the initial training phase, with YOLOv11 and YOLOv12 achieving higher and more stable peak values close to 1.0 than earlier versions. The chart in Figure 6 shows a steep increase in detection accuracy during early epochs for all models, with the YOLOv11 and YOLOv12 series achieving higher and more stable peak mAP scores close to 1.0. The YOLOv10 variants show comparatively lower final mAP, indicating architectural and algorithmic improvements in the newer versions. These results highlight the progressive enhancements in detection capabilities and real-time performance achievable with evolving YOLO architectures.

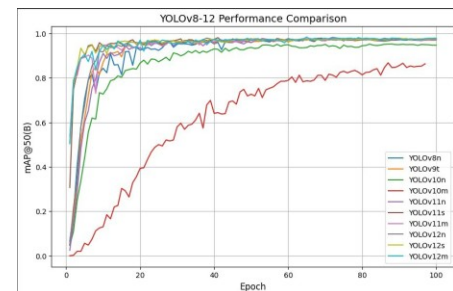


Fig. 6. Comparative mAP@50 performance of YOLOv8 to YOLOv12 variants over 100 training epochs.

3. Precision-Recall Curve: The precision-recall (PR) curve for all classes is presented in Figure 7. The curve, which

TABLE II
COMPUTATIONAL COMPLEXITY AND INFERENCE PERFORMANCE COMPARISON OF YOLOv10, YOLOv11, AND YOLOv12 VARIANTS

Model	Params (M)	GFLOPs	Model Size (MB)	Latency (ms)	FPS
YOLOv10-n	2.72	8.48	5.52	12.23	81.75
YOLOv10-s	8.10	24.95	15.83	12.00	83.36
YOLOv10-m	16.53	64.23	32.02	14.83	67.44
YOLOv11-n	2.60	6.48	5.24	12.52	79.86
YOLOv11-s	9.44	21.64	18.33	13.19	75.83
YOLOv11-m	20.09	68.36	38.71	15.21	65.76
YOLOv12-n	2.58	6.50	5.29	17.63	56.72
YOLOv12-s	9.27	21.61	18.09	17.67	56.60
YOLOv12-m	20.17	67.92	38.94	19.30	51.82

reports an $mAP@0.5$ of 0.981, is sharply skewed toward the top-right, signifying near-perfect precision and recall with extremely low error rates. The minimal drop-off before recall reaches 1.0 emphasizes the efficiency of the current YOLO architecture on this application. The curve's proximity to the top-right corner indicates high precision and recall values, reflecting excellent detection performance with minimal false positives and false negatives. This demonstrates the model's strong capability in correctly identifying retail items, effectively balancing sensitivity and specificity.

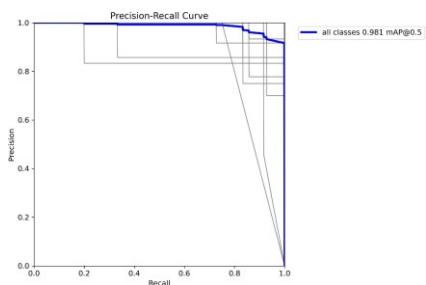


Fig. 7. Precision-Recall (PR) curve illustrating the trade-off between precision and recall for all classes with an average $mAP@0.5$ of 0.981.

E. Real-World Detection Results

To further determine the reliability and repeatability of the experimental results, as well as their stability, three experiments using YOLO models were carried out independently under identical conditions and with the same dataset split each time. The results showed only a small range of variation. The Average Precision ($mAP@50$) varied within a range of $\pm 0.3\%$, and the change in inference latency was well below ± 1 ms. From these observations, it can be concluded that the performance of the YOLO models is stable and reliable and is not significantly affected by experimental variability. As shown in Figure 8, the YOLOv12m model demonstrates a robust capability to accurately identify a variety of grocery items, including chocolates, cleaning products, and snacks, under varied backgrounds and lighting conditions.

V. CONCLUSION

This study successfully implemented and evaluated various YOLO models (YOLOv10, YOLOv11, and YOLOv12 with their nano, small, and medium variants) for a real-world retail



Fig. 8. Detection results on a broader retail product assortment, showcasing accurate identification of diverse grocery items using YOLOv12.

self-checkout system. The comparative analysis demonstrated the superior performance of the newer YOLO architectures on a custom dataset of diverse Indian retail products. The project's primary objective was to identify the most effective model for this task, and the results clearly show that the **YOLOv12m** model achieved the highest performance metrics, including a commendable $mAP@50$ score of 98.13%. This high level of accuracy and precision is critical for the reliable and efficient functioning of an automated self-checkout system, significantly reducing the chances of misidentification and errors. The findings affirm that these advanced models, when trained on a relevant dataset, are highly suitable for complex object detection tasks in a dynamic retail environment.

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